PREP 1 · LESSONS 1–2

Stores and the kinetic store

Energy stores and transfers · describing a transfer · kinetic energy

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define precision. DN 850 [1]

2 What is a control variable? DN 872 [1]

3 What is an anomalous result? DN 895 [1]

Key ideas LESSONS 1–2

Show your working

4 A cyclist brakes and stops. Complete the sentence.

Energy is transferred _____ **(mechanically / electrically / by radiation)** from the _____ **(kinetic / chemical / thermal)** store of the cyclist to the _____ **(elastic / thermal / nuclear)** store of the brakes. [2]

5 A ball of mass 0.50 kg moves at 4.0 m/s. Use $E_k = \frac{1}{2} m v^2$

Calculate the energy in the kinetic store of the ball. [3]

6 Name the two quantities that kinetic energy depends on. [1]

7 A car's speed doubles. Its kinetic energy becomes **2 times** | **4 times** | **8 times** as big. [1]